

Justification and Approval (J&A) for Other Than Full and Open Competition

Was a J&A approved for the preceding acquisition where that acquisition required a J&A? ☒ Yes ☐ No

!! Attach the preceding J&A in the staff package for this J&A. The preceding J&A will be used as a reference document.

Is this a new or amended J&A Document? ☒ New ☐ Amended (Prior to Award Only!)

Is this a Bridge Action as defined at [AFFARS 5302.101](#)? ☐ Yes ☒ No

Dollar Value of this Acquisition: ☐ ≤ \$750K ☒ > \$750K and ≤ \$15M ☐ > \$15M and ≤ \$100M ☐ > \$100M

Contracting Activity: United States Air Force; Air Force Installation Contracting Command; 21 CONS/PKA

Purchase Request (if available) / Local ID Number: N/A

Program / Project (and PE, if applicable): Johnson Controls Equipment and Services

Program Type (PEO, Enterprise, of Operational): Operational

Authority:

6.302-1 – 10 USC 3204(a)(1), Only One Responsible Source - No Other Supplies or Services Will Satisfy Agency Requirements

Estimated Contract Cost (including options):

J&A Type: ☒ [Class](#) ☐ Individual

!! Provide estimated cost of all contracts.

COORDINATION ([AFFARS 5306.304\(a\)](#))

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I. Agency and Contracting Activity.

United States Air Force; Air Force Installation Contracting Command; 21 CONS/PKA
135 Dover ST BLDG 350 STE 1055
PETERSON SFB, CO 80914-1192

II. Nature and/or description of the action being approved.

_____ has a reoccurring requirement for Johnson Controls, Inc. (JCI) parts, equipment, services and training to maintain and repair the Environmental Management Control Systems (EMCS) Network at Peterson Space Force Base (PSFB) and Cheyenne Mountain Space Force Station (CMSFS). Future contract actions will be driven by EMCS requirements consistent with the _____ 10-year plan for maintaining the EMCS systems as further defined in Section III. The Government anticipates firm fixed price (FFP) orders against JCI's GSA Schedule #GS-07F-190CA for parts, and FFP purchase orders directly awarded to JCI for services and training. The action will be for new contracts.

III. Description of supplies/services required to meet agency needs.

_____ has a requirement to procure JCI parts, equipment, maintenance and training for their EMCS Network. _____ has a 10-year plan to maintain the ECMS Network, as doing anything less than maintaining it is to the detriment of the Air Force and operations at PSFB and CMSFS. Only JCI parts and services can be utilized in the existing networks due to system interoperability and posterity, and non-utilization of JCI parts and services risks voiding existing JCI system warranties.

_____ anticipates three non-scheduled maintenance requirements each year with an average cost of _____ which will total to _____. Additionally, _____ based on availability of funds. Forecasted out, the government expects at least _____

Furthermore, _____ expects there to be an additional preventative maintenance cost for the JCI EMCS components on PSFB. This is estimated to be _____

_____ used seven (7) quotes they received from JCI for parts, services, and training requirements alongside personal knowledge of the systems and service to come up with their estimate for dollar amount and number of parts to procure. These seven (7) quotes represent part of _____ projected future projects within their 10 year plan, and adequately shed light on what _____ expects for annual requirements. One such requirement is the upgrade of the Direct Digital Control (DDC) Devices. This upgrade consists of replacing and upgrading all legacy DDC Devices in certain buildings with newer generation JCI BACnet Master Slave Token Passing (MSTP) compatible controls. These controls shall also be mapped and pointed to the main EMCS Metasys server located in _____ with updated Metasys User Interface (MUI) graphics on the EMCS front end.

Over the 10 years covered by this J&A, _____ has annually targeted DDC upgrades _____ based on facility system operations and technology product end of life support. Individual facilities are targeted and assesses annually as funding becomes available. Costs for building upgrades are directly derived from the number of controllers present, but, as previously stated, estimated to average around _____

Over the 10 years covered by this J&A, parts, equipment and service acquisitions for unit/system maintenance, repairs, and training service are estimated to be _____. The Government finds it in its best interest to have this Class J&A have a ceiling of _____ to account for increases in price over the next 10 years due to myriad factors including, but not limited to, inflation and supply chain pressure. This forecasted _____ is only for the existing system; a full system replacement is outside the scope of this class waiver.

There is no intellectual property associated with requirements associated with this Class J&A.

IV. Demonstration that the contractor's unique qualifications or the nature of the acquisition requires use of the authority cited above.

The Peterson EMCS Network and unclassified DDC supervisory devices consists of a standardized JCI Metasys platform of _____

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materials and equipment. Changing equipment platforms would create additional costs in training and slower system recovery times as technicians will not be familiar with external component troubleshooting. Due to the existing types of supervisory panels, DDC field components and network configuration, JCI components are required to ensure compatibility, security, interoperability and certification within the existing systems. A large portion of the network is currently dedicated to EMCS, which provide remote monitoring and access for HVAC shop to the majority of facilities on base.

utilize JCI. All of these buildings' EMCS currently The buildings managed by

have been reliant on JCI for their EMCS programs for such an extended period of time that working with any other vendor is inconceivable as JCI systems are absolutely entrenched on PSFB. Therefore, IAW FAR 6.302-1, there is only one responsible source.

The reliance on JCI stems from the fact that attempting to integrate different services and equipment brands into these EMCS protocols creates instability and loss of function in the network. Furthermore, other vendor application owners use customized configurations without considering the impact on the entire IT environment and existing systems; they are not necessarily backward compatible. As a result, a mix of different technologies can result in significant communication issues and command capabilities, such as not responding to system operator issued commands. That, combined with old and failing equipment, will cause frequent outages and wasted resources—an unacceptable risk for PSFB/CMSFS and their highly important mission sets.

already has an EMCS service contract to provide mission critical support to a network with 97% of the systems already configured and sourced with JCI parts and logic. This contract was a sole source award to JCI and utilized a Limited Sources Justification JCI warranties are also managed within the aforementioned existing service contract. Parts are also currently purchased with this service contract. It is anticipated that this requirement will continue under this class J&A as outlined in section III. Additionally, parts can be procured project-wide with individual, total hardware upgrades such as main Network Automated Engines (NAE) as well as during entire facility DDC upgrades. is not able to do this with other vendors nor would it be cost effective given the limited scope other equipment providers have on EMCS network. Furthermore, these EMCS systems are proprietary technology to JCI. As such, only JCI certified technicians are authorized to work on these systems and provide training. This J&A will be used to support the acquisition of a new Operations and Maintenance (O&M) contract supporting the day to day operations of the EMCS.

Replacement of the current systems is not a viable option at this time given the anticipated life of the existing system is at least 10 years and would cost an estimated to replace the current EMCS system. Moreover, the EMCS system would take an estimated 3 months to recover once parts are acquired. This is an unacceptable delay given how the mission would be affected. would lose remote functionality of their facility industrial equipment throughout the installation—an implausible trade-off. would also lose the ability of their remote alarm notifications system that alerts shop personnel of critical failures within HVAC, sewage lift stations and emergency power generation systems. These systems monitor and control facility cooling and heating equipment that often support mission critical infrastructure within base facilities. Chillers, boilers and air delivery systems provide conditioned air to critical IT spaces and equipment. The EMCS system has automated features that alert stand-by personnel of issues needing immediate attention and response before massive failures occur. This system allows shop personnel to react in a proactive measure. It is critical that the system communication and control features are available for operational use and have dependable integrity within its communications to the user operator work stations within the network. It is likely that award to any other source would result in substantial duplication of cost to the Government that is not expected to be recovered through competition. Furthermore, non-interoperability among EMCS system providers would result in a similar issue if this were to be competed.

V. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by subpart 5.2 and, if not, which exception under FAR 5.202 applies.

JCI has business in the commercial and residential market with authorized dealers are only able to conduct business in their residential market. the Government is exclusively a commercial customer and therefore unable to use any authorized dealers. All commercial parts, maintenance, and trainings are handled entirely and solely by JCI themselves. With the Government as a commercial customer it is inefficient to solicit requirements related to JCI anywhere besides JCI directly; JCI is the only source capable of supporting the requirements. This was then further confirmed after speaking with JCI representatives. JCI's vertically integrated supply chain directly controls every aspect of their commercial business. JCI,

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however, will utilize other specialty businesses (i.e. crane companies or electricians) to complete specific maintenance jobs when it is needed.

The Government posted a sources sought on GSA and SAM.gov on 10 August 2022. No vendors responded to the sources sought as being capable and interested in competing for these requirements.

VI. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

Each contemplated purchase for parts or service will be required to have an individual determination of price fair and reasonableness IAW FAR 13.106-3(a) or FAR 15.403-3(c)(1) prior to award. As such, contracts supporting through this J&A will rely on myriad pricing techniques, including comparison of current contract pricing for similar requirements currently being used by DOD. will continue to conduct market research through Internet searches, published catalogs and/or GSA schedules of similar characteristics. Based upon all of the above, the contracting officer expects they will be able to determine that the anticipated cost to the Government will be fair and reasonable.

VII. Description of the market research conducted and the results, or a statement of the reasons market research was not conducted.

The government researched JCI, their supply chains, and their business practices by speaking with representatives from the company, speaking with government personnel familiar with JCI, and by using Internet searches. The information gathered from these market research efforts concluded that certified JCI parts and services, for commercial customers, can only be obtained directly through JCI.

will continue to conduct market research to the maximum extent practicable to support contract actions awarded via this Class J&A in the procurement of JCI parts and/or services.

VIII. Any other facts supporting the use of Other Than Full and Open Competition.

For the existing EMCS system to communicate to other products, it would need to work through the JCI "Metasys" platform that exist on Peterson Space Force Base. The JCI Metasys system is the main nucleus and dashboard for the entire EMCS system that is controlled by All devices on the EMCS network report to the JCI Metasys server and all users log into it for system management. JCI Metasys is capable of communicating to all of the major industrial control systems currently at Peterson. JCI Metasys currently integrates system components that allow protocols such as MODBUS and BACnet to be integrated. These are the primary protocols used throughout the building automation industry. The problem is that other similar vendor products often do not authorize full integration to their product data and control algorithms. In order to get this kind of functionality to perform troubleshooting and maintenance, the customer has to purchase extra vendor specific proprietary software tools and servers. This would also mean that vendor specific training would be required by the base maintenance shops for each specific vendor system. This also often results in data communication issues and failures. Items from different vendors have been found to share their data at different rates resulting in intermittent communication issues and often become unreliable. So, other products can communicate their data to a point, but they do not allow the top to bottom functionality to fully perform the kind of routine maintenance and troubleshooting required without buying their extra software. Furthermore, dependence on JCI and its repeated focus in their acquisition of building automation over the years has resulted in a greatly entrenched JCI set up throughout Peterson Space Force Base. To acquire other vendor's hardware or software would be a detriment to the mission of

IX. List of any sources that expressed, in writing, an interest in the acquisition.

See Section VI above.

X. A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisitions for the supplies or services required.

Replacing the EMCS would create unnecessary additional costs in training and slower system recovery times as technicians

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will not be familiar with external component troubleshooting. As roughly of the buildings on Peterson Space Force Base that are managed by use JCI systems, replacing them outright is not feasible in the foreseeable future. Integrating different services and equipment brands that are not JCI parts creates instability and loss of function in the network. Also, mixing platforms and protocols will result in very little standardization, creating system integration and daily maintenance challenges. Therefore, at this time or in the near future, no actions will be made to replace the EMCS parts with other similar parts in future acquisitions.

As the J&A approaches expiration, will reevaluate the price and downtime caused by replacing the JCI systems to see if these system replacements would be beneficial and in the Government's best interest.

XI. Certification by the Contracting Officer.

As evidenced by my signature above, I certify that this justification is accurate and complete to the best of my knowledge and belief.

XII. Certification by the technical/requirements personnel.

As evidenced by my signature above, I certify that any supporting data contained herein, which is my responsibility, is both accurate and complete.